

IMMUNOLOGY

Altered homeostasis and development of regulatory T cell subsets represent an IL-2R–dependent risk for diabetes in NOD mice

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The cytokine interleukin-2 (IL-2) is critical for the functions of regulatory T cells (T_{regs}). The contribution of polymorphisms in the gene encoding the IL-2 receptor α subunit (*IL2RA*), which are associated with type 1 diabetes, is difficult to determine because autoimmunity depends on variations in multiple genes, where the contribution of any one gene product is small. We investigated the mechanisms whereby a modest reduction in IL-2R signaling selectively in T lymphocytes influenced the development of diabetes in the NOD mouse model. The sensitivity of IL-2R signaling was reduced by about two- to threefold in T_{regs} from mice that coexpressed wild-type IL-2R β and a mutant subunit (IL-2R β ^{Y3}) with reduced signaling (designated NOD-Y3). Male and female NOD-Y3 mice exhibited accelerated diabetes onset due to intrinsic effects on multiple activities in T_{regs}. Bone marrow chimera and adoptive transfer experiments demonstrated that IL-2R β ^{Y3} T_{regs} resulted in impaired homeostasis of lymphoid-residing central T_{regs} and inefficient development of highly activated effector T_{regs} and that they were less suppressive. Pancreatic IL-2R β ^{Y3} T_{regs} showed impaired development into IL-10–secreting effector T_{regs}. The pancreatic lymph nodes and pancreases of NOD-Y3 mice had increased numbers of antigen-experienced CD4⁺ effector T cells, which was largely due to impaired T_{regs}, because adoptively transferred pancreatic autoantigen–specific CD4⁺ Foxp3[–] T cells from NOD-Y3 mice did not accelerate diabetes in NOD.SCID recipients. Our study indicates that the primary defect associated with chronic, mildly reduced IL-2R signaling is due to impaired T_{regs} that cannot effectively produce and maintain highly functional tissue-seeking effector T_{reg} subsets.

INTRODUCTION

CD4⁺ Foxp3⁺ regulatory T cells (T_{regs}) maintain peripheral tolerance by suppressing autoreactive T cells that escape negative selection in the thymus. Interleukin-2 (IL-2), through its interaction with the high-affinity IL-2 receptor (IL-2R), which consists of IL-2R α (CD25), IL-2R β (CD122), and γ_c (CD132), controls multiple critical activities in T_{regs} (1–4). IL-2R signaling is essential for thymic T_{reg} development, promoting the maturation of CD4⁺ CD25^{lo} Foxp3^{lo} immature cells into functional T_{regs} (5–9). IL-2 also contributes to T_{reg} peripheral homeostasis (10), in part by providing survival signals to central or resting CD62L^{hi} T_{regs} (11) and by supporting the development of terminally differentiated, short-lived Klrp1⁺ T_{regs} (12). T_{reg} identity and suppressive function are enforced by IL-2 (13) through direct regulation of the expression of *Foxp3*. This property also influences the development of conventional T cells into peripherally induced T_{regs} (14–16). Because *Foxp3* represses the expression of *IL2*, T_{regs} constitutively express the high-affinity trimeric IL-2R to respond to IL-2 produced by effector T (Teff) cells and, in some cases, dendritic cells (17, 18).

Multiple single-nucleotide polymorphisms (SNPs) in *IL2*, *IL2RA*, and *IL2RB* are associated with an increased risk for developing several human autoimmune diseases (19). On their own, these IL-2–related SNPs, some of which are common in the population, represent a small risk for autoimmunity and most likely act in concert with SNPs in other genes as well as environmental factors to trigger

autoimmunity. This complexity has made it difficult to determine how an individual SNP promotes autoimmunity. With respect to type 1 diabetes (T1D), individuals with susceptible SNPs in *IL2RA* (20) have a reduced abundance of CD25 on T_{reg} and T memory cells, which leads to reduced IL-2R signaling (21). Some data have also associated reduced IL-2R signaling in T_{regs} with decreased fitness and suppressive function (22, 23). Nevertheless, we still poorly understand how a subtle reduction in IL-2R signaling represents a risk for autoimmunity, including T1D.

The nonobese diabetic (NOD) mouse has been widely used as a model for T1D, where *Il2* is a genetic risk for development of diabetes. The importance of the insulin-dependent diabetes risk locus 3 (*Idd3*), which contains *Il2*, for diabetes is illustrated by the substantial reduction in diabetes occurrence in NOD congenic mice in which the *Idd3* interval is derived from C57BL/6 mice (24). NOD *Idd3* results in a twofold reduction in IL-2 production by CD4⁺ T cells (25). Moreover, *Il2* mRNA was selectively reduced in infiltrating cells in pancreatic islets of NOD mice when compared to that in peripheral immune tissues (26). Reduced IL-2 production is associated with a pancreas-specific decrease in the T_{reg} to Teff cell ratio, which might reflect impaired T_{reg} homeostasis. Pancreatic T_{regs} in NOD mice have reduced amounts of CD25 and Bcl-2 but increased Ki67 abundance, the latter of which may reflect a compensatory proliferative response toward autoreactive T cells (26, 27). These studies concluded that the *Il2*-related genetic risk for diabetes in NOD mice is primarily due to a pancreas-specific impairment of T_{reg} homeostasis that reduced the ability of these cells to suppress autoreactive T cells. However, these findings indirectly support these conclusions, and the effect on IL-2R signaling as a risk for diabetes remains poorly understood. An additional complication in clearly defining the *Il2*-related risk for diabetes in NOD mice is that the *Idd3* locus also

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contains *Il21*, which is closely linked to *Il2*. This polymorphism in *Il21* results in increased secretion of IL-21, a proinflammatory cytokine, and this increase in IL-21 abundance is linked to diabetes susceptibility in NOD mice (28).

As discussed earlier, the direct contribution of the *Il2*-related risk for diabetes in NOD mice remains poorly understood. In addition, the reduced IL-2 production associated with the *Idd3* locus does not directly test the consequences of altered IL-2R signaling, which is a risk for T1D and several other autoimmune diseases. To directly target IL-2R signaling, one must affect the activity of IL-2R β , because this subunit is responsible for the distinct signaling attributed to IL-2. Simply knocking out *Il2rb* in the germ line or selectively in T_{regs} leads to the production of immature, nonfunctional T_{regs}, which results in rapid lethal systemic autoimmunity (29–31); this approach is not suitable to assess how subtle changes in IL-2R signaling might promote autoimmunity. For these reasons, we developed a model in which IL-2R β signaling was selectively reduced in all T cells of NOD mice. We reasoned that IL-2R-dependent processes in T cells relevant to diabetes development in NOD mice would be intensified and thus distinguished from other genetic risks in this model. Diabetes was accelerated in male and female NOD mice in which IL-2R β signaling was modestly and selectively reduced in T cells. Furthermore, this autoimmunity was directly related to substantial changes in T_{regs} that included, but were not limited to, altered homeostasis and function, whereas more modest alterations were noted in the Teff compartment.

RESULTS

Expression of the IL-2R β^{Y3} transgene in NOD mice reduces IL-2R signaling

We previously expressed a mutant IL-2R β , designated IL-2R β^{Y3} , in mice such that three critical cytoplasmic tyrosine residues, Tyr³⁴¹, Tyr³⁹⁵, and Tyr⁴⁹⁸, were mutated to phenylalanines (Y341F, Y395F, and Y498F) (32). These mutations interfere with the association of the adaptor Shc and transcriptional regulator signal transducer and activator of transcription 5 (STAT5) to the cytoplasmic tail of IL-2R β and thus substantially reduce IL-2R signaling. IL-2R β^{Y3} was targeted for exclusive expression by T lymphocytes using the *Cd2* minigene. When crossed onto the *IL-2R $\beta^{-/-}$* genetic background of C57BL/6 mice, the reduced IL-2R β signaling associated with IL-2R β^{Y3} outwardly supported normal T_{reg} development, homeostasis, and function, whereas Teff and T memory compartments were much more obviously impaired (32). However, in a competitive environment, IL-2R β^{Y3} T_{regs} did not compete with wild-type (WT) T_{regs}, demonstrating that IL-2R β^{Y3} T_{regs} are not fully functional when IL-2R β signaling is impaired (7).

The availability of mice that express IL-2R β^{Y3} under the control of the *Cd2* promoter, designated as Y3 mice, provided an opportunity to directly model how impaired IL-2R β signaling in T cells represents a risk for autoimmunity. Here, we tested this concept for diabetes development by reducing IL-2R β signaling in conventional T cells and T_{regs} in NOD mice. C57BL/6-Y3 *IL-2R $\beta^{-/-}$* mice were backcrossed to NOD *IL-2R $\beta^{+/+}$* mice for 12 generations to derive NOD-Y3 *IL-2R $\beta^{+/+}$* and NOD-Y3 *IL-2R $\beta^{+/-}$* mice. SNP analysis of NOD-Y3 mice indicated that all chromosomal regions, including all *Idd* loci, were of NOD origin except for a region within chromosome 15 that surrounds *Il2rb*. We anticipated that, when the IL-2R β^{Y3} transgene was expressed in NOD mice on the *IL-2R $\beta^{+/+}$* or *IL-2R $\beta^{+/-}$* genetic background, the competition for IL-2 between WT IL-2R β and IL-2R β^{Y3} would lead to an overall reduction in IL-2R β signaling. Thus, all studies

in this report (except when specified otherwise) used mice for which the IL-2R β^{Y3} transgene was crossed onto *IL-2R $\beta^{+/+}$* or *IL-2R $\beta^{+/-}$* NOD mice. As expected, IL-2R β^{Y3} was detected on essentially all T lymphocytes (Fig. 1A), as indicated by the increased mean fluorescence intensity (MFI) on T_{reg} and CD8⁺ T cells, which also had endogenous WT IL-2R β , and by essentially all conventional CD4⁺ T cells, which were mostly IL-2R β^{neg} in normal NOD mice. In contrast, the MFI of CD25 was reduced by 20 to 30% for T_{regs} in NOD-Y3 mice for all examined tissues (Fig. 1B). Because IL-2 increases the abundance of CD25 on the cell surface (33), this result suggests that IL-2R signaling is reduced in T_{regs} from NOD-Y3 mice.

Dose-response studies that assessed the tyrosine phosphorylation of STAT5 (pSTAT5) showed that T_{regs} expressing IL-2R β^{Y3} had a reduced response to IL-2 (Fig. 1C). The median effective concentration (EC₅₀) value of IL-2 for WT T_{regs} was 3.1 pM, whereas the EC₅₀ values for IL-2R β^{Y3} T_{regs} on the *IL-2R $\beta^{+/+}$* and *IL-2R $\beta^{-/-}$* background were 8.2 and 26.8 pM, respectively. Thus, T_{regs} that coexpressed the WT IL-2R β and IL-2R β^{Y3} were 2.6-fold less responsive to IL-2 than were WT T_{regs} (Fig. 1C). In addition, the response by T_{regs} to IL-2 was impaired in NOD-Y3 mice, as demonstrated by the decreased percentage of NOD-Y3 cells that were pSTAT5⁺ (Fig. 1D). The reduced activation of STAT5 could be attributed to IL-2, because treatment of the mice with anti-IL-2 antibody for 24 hours substantially reduced the percentage of pSTAT5⁺ cells (Fig. 1D). Minimal IL-2-dependent STAT5 phosphorylation was stimulated in conventional CD4⁺ and CD8⁺ T cells from NOD and NOD-Y3 mice when assessed directly ex vivo or after in vitro challenge with IL-2, which was due to the lack of the high-affinity IL-2R on these cells (fig. S1, A and B). To determine whether IL-2R β^{Y3} altered IL-2R β signaling in Teff cells, IL-2-dependent pSTAT5 abundance was assessed for T cells preactivated with anti-CD3 and anti-CD28 (fig. S1, C and D). IL-2 signaling sensitivity was not altered by IL-2R β^{Y3} under these conditions because these cells had high amounts of all WT IL-2R subunits that likely out-compete the mutant IL-2R β^{Y3} for IL-2. Nevertheless, because Teff cells in vivo transiently express the IL-2R at amounts typically lower than those of T_{regs} (34), we expected that IL-2R β^{Y3} would also decrease IL-2R signaling in these cells by at least two to threefold as was detected for T_{regs}.

IL-2R β is also a subunit of the IL-15R. Thus, IL-2R β^{Y3} is also expected to affect IL-15-dependent signaling. IL-2R β^{Y3} expression lowered IL-15-induced signaling in T_{regs} and CD8⁺ T cells by about 2.9- and 2.3-fold, respectively (Fig. 1E). This EC₅₀ shift was similar to that noted for IL-2 on T_{regs} between NOD-Y3 and WT NOD mice (Fig. 1C). Because IL-2R β was not detected in CD4⁺ Foxp3[−] T cells, the response to IL-15 in NOD-Y3 mice was directly mediated by IL-2R β^{Y3} . The EC₅₀ for CD4⁺ Foxp3[−] T cells from NOD-Y3 mice was 100 pM, which is about 10-fold higher than that measured for WT T_{regs} or CD8⁺ T cells. This high EC₅₀ implies that IL-2R β^{Y3} CD4⁺ Foxp3[−] T cells would only respond to IL-15 in environments that have a high concentration of this cytokine. Thus, IL-2R β^{Y3} is more likely to potentially affect the responses of T_{regs} and CD8⁺ T cells to IL-15 than to affect those of conventional CD4⁺ T cells. This is an unavoidable potential off-target consequence of directly interfering with IL-2R β signaling.

IL-2R β^{Y3} accelerates the onset and increases the penetrance of diabetes in NOD-Y3 mice

IL-2R β^{Y3} acted in a dominant-negative manner to reduce IL-2R signaling by two- to threefold in NOD-Y3 mice. Therefore, we assessed the consequences of this modest reduction in IL-2R signaling by

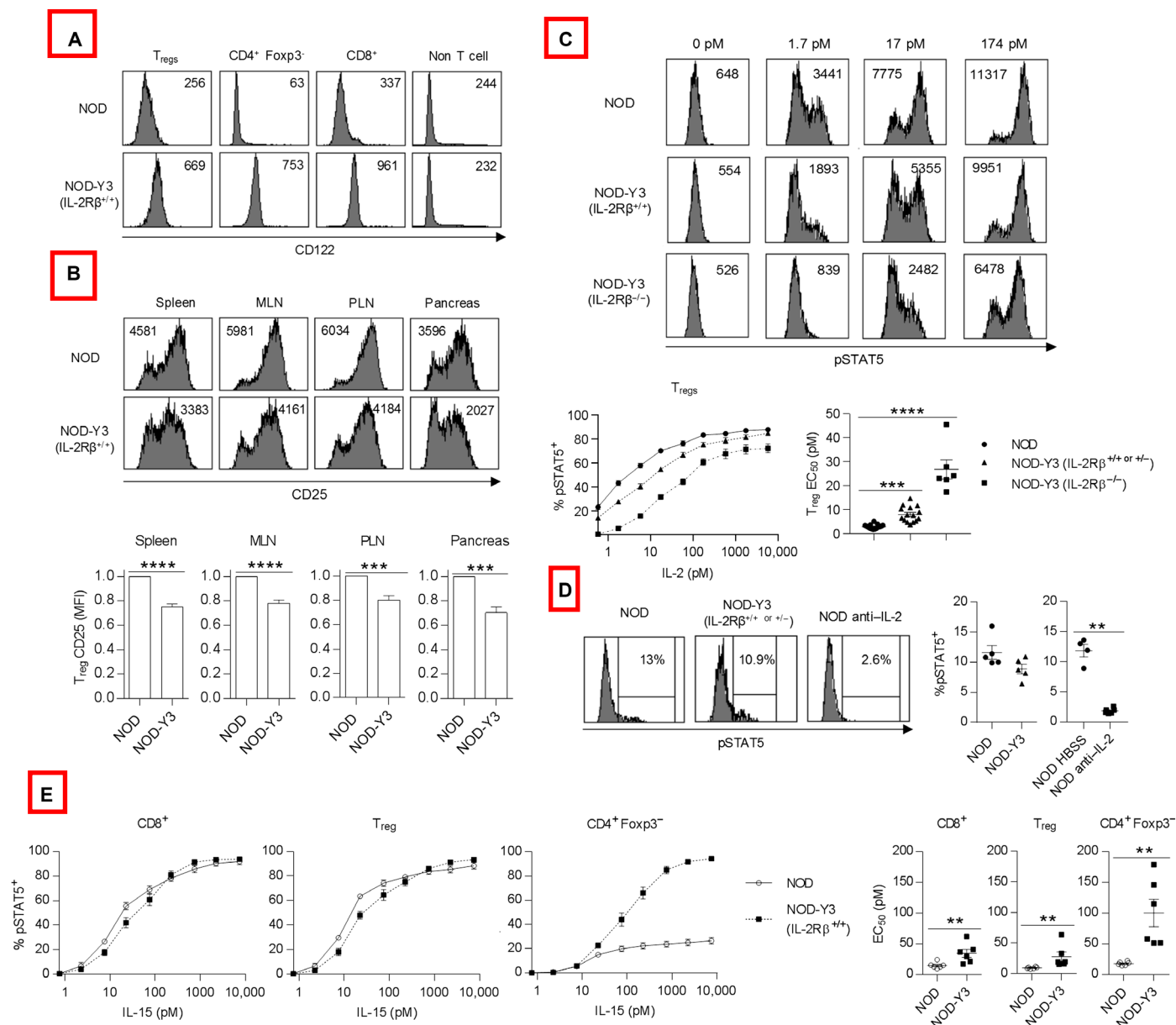


Fig. 1. Analysis of IL-2R abundance and STAT5 activation in lymphoid cells from NOD and NOD-Y3 mice. (A and B) Flow cytometry analysis of the cell surface expression of CD122 (A) and CD25 (B) on the indicated cells from the spleens of the indicated mice ($n = 15$ to 18 mice). NOD, nonobese diabetic. (B) Bottom: The mean fluorescence intensity (MFI) for CD25 expression by regulatory T cells (T_{regs}) was assessed by the Wilcoxon signed-rank test. MLN, mesenteric lymph node; PLN, pancreatic lymph node. (C) Analysis of pSTAT5 abundance in CD4⁺ Foxp3⁻-gated T_{regs} after stimulation with the indicated concentrations of interleukin-2 (IL-2) for 15 min in vitro. Top: Representative flow cytometry plots with pSTAT5 MFI. Bottom: Dose-response curves (left), which were subjected to nonlinear regression analysis to determine the median effective concentration (EC₅₀) values (right). Data are from NOD IL-2Rβ^{+/+} or NOD IL-2Rβ^{-/-} ($n = 15$), NOD-Y3 IL-2Rβ^{+/+} or NOD-Y3 IL-2Rβ^{-/-} ($n = 15$), and NOD-Y3 IL-2Rβ^{-/-} ($n = 6$) mice and the resulting EC₅₀ values were analyzed by the Kruskal-Wallis test. (D) Flow cytometry analysis of pSTAT5 abundance in T_{regs} directly after isolation from the indicated mice ($n = 5$). In some cases, the NOD mice were treated with Hanks' balanced salt solution (HBSS) ($n = 4$) or anti-IL-2 ($n = 6$) for 24 hours before the cells were isolated and analyzed for pSTAT5. The resulting data were analyzed by Mann-Whitney test. (E) Flow cytometry analysis of pSTAT5 abundance in the indicated T cell populations stimulated in vitro with the indicated concentrations of IL-15 for 15 min. Dose-response curves (left) were subjected to nonlinear regression analysis to determine the EC₅₀ values (right), which were analyzed by the Mann-Whitney test ($n = 6$). ** $P < 0.01$, *** $P < 0.001$, **** $P < 0.0001$.

comparing diabetes onset in male and female NOD and NOD-Y3 mice (Fig. 2A) and NOD and NOD-Y3 BDC2.5 T cell receptor (TCR) transgenic mice (Fig. 2B), whose CD4⁺ T cells express a TCR specific for the chromogranin A autoantigen. In male mice, the average time to diabetes onset was 23 weeks for NOD-Y3 mice but was greater than 40 weeks for WT NOD mice. In female mice, the average diabetes onset was 18 weeks for NOD-Y3 mice but was 27 weeks for WT NOD

mice (Fig. 2A). Similarly, average diabetes onset was accelerated in both male and female NOD-Y3 BDC2.5 mice (Fig. 2B). Because of the unexpected break in tolerance in male mice, most of our further studies focused on male NOD and NOD-Y3 mice.

In polyclonal mice, serum insulin autoantibody titers were also greater in male and female NOD-Y3 mice than in NOD mice when measured at 12 weeks of age (Fig. 2C). The overall cell numbers recovered

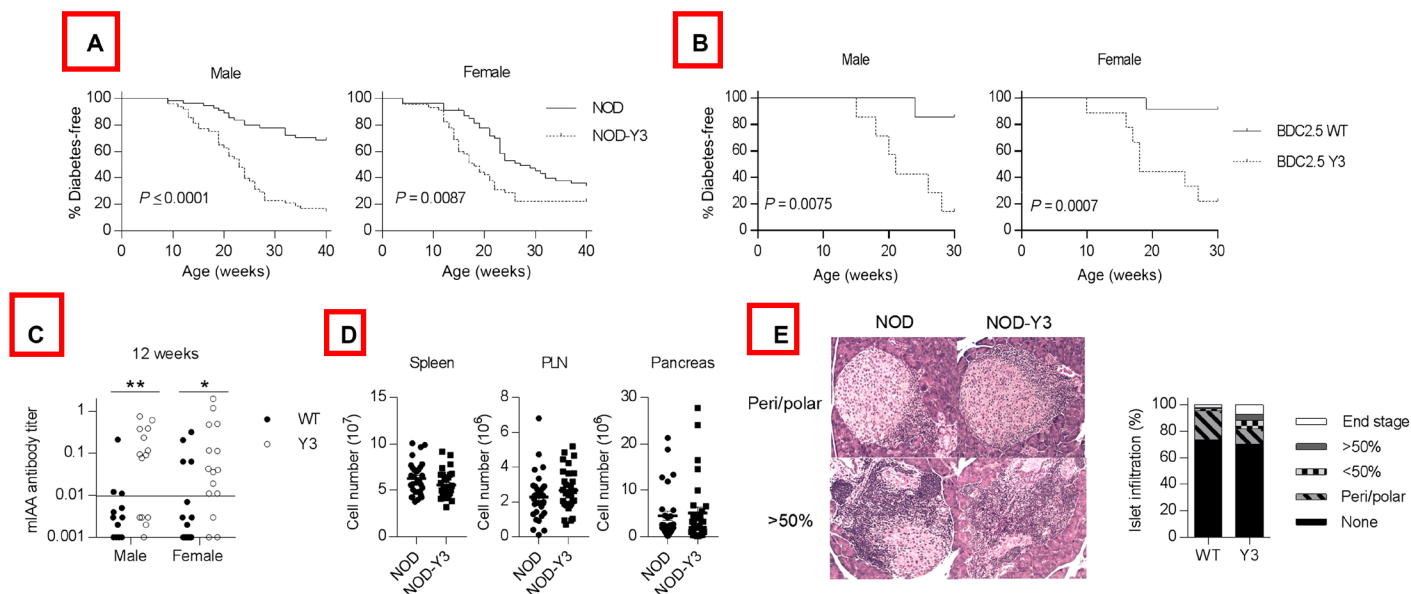


Fig. 2. The effect of IL-2R β^{Y3} on diabetes in NOD mice. (A and B) Diabetes onset was followed in male and female NOD and NOD-Y3 mice ($n = 45$ to 54 mice) for up to 40 weeks of age (A) and in male ($n = 7$) and female ($n = 9$ to 12) NOD BDC2.5 and NOD-Y3 BDC2.5 mice for up to 30 weeks of age (B). Data were analyzed by log-rank test, and the resulting EC₅₀ values were analyzed by Mann-Whitney test. (C and D) Insulin autoantibody titers (mIAA) from male and female NOD and NOD-Y3 mice at 12 weeks of age (C) and analysis of cellularity of the indicated tissues from male mice ($n = 30$) 8 to 16 weeks of age (D). Data were analyzed by Mann-Whitney test. (E) Pancreatic islet infiltration scoring by hematoxylin and eosin staining of pancreas sections from male mice at 7 to 12 weeks of age (>200 islets were evaluated). * $P < 0.05$, ** $P < 0.01$.

from the pancreatic lymph node (PLN) and the pancreas were slightly, but not significantly, increased in male NOD-Y3 mice compared to those in male NOD mice (Fig. 2D). Male NOD and NOD-Y3 mice also had a similar percentage of islets showing lymphocytic infiltration; however, NOD-Y3 mice had more late-stage infiltration compared to that of the WT controls (Fig. 2E). These data suggest that lymphoid cells present in the pancreas were somewhat more effective in attacking islet tissues in the NOD-Y3 pancreas and likely contributed to the acceleration of diabetes onset.

IL-2R β^{Y3} alters T_{regs} and antigen-experienced T cells in NOD mice

Although the overall cellularity in male NOD-Y3 mice was relatively normal, the T lymphocyte composition was distinctive in lymphoid tissues and the pancreas. There were less T_{regs} in the spleen, PLN, and pancreas of NOD-Y3 mice compared to those of NOD mice (Fig. 3A). These T_{regs} exhibited a slight but statistically significant reduction in Foxp3 abundance together with a reduced amount of CD25 (Fig. 1B). Conventional CD4⁺ T cells were slightly increased in number in the spleen and PLN, whereas antigen-experienced CD4⁺ Foxp3⁺ CD44^{hi} T cells were increased in number in the PLN and pancreas (Fig. 3B). In contrast, a reduced fraction of CD8⁺ T cells was noted for these tissues, whereas the proportion of antigen-experienced CD8⁺ CD44^{hi} T cells remained largely unaltered (Fig. 3C). Most of these changes were also noted for female NOD-Y3 mice (fig. S2). Two exceptions included a less marked effect on T_{reg} and CD4⁺ Foxp3⁺ CD44^{hi} percentages in the pancreas of female NOD and NOD-Y3 mice. This result is most likely because inflammation leading to diabetes is normally more severe in female NOD mice than in male NOD mice, which might have obscured the effects of IL-2R β^{Y3} . For NOD-Y3 BDC2.5 CD4⁺ T cells, reducing IL-2R β signaling resulted in a consistent trend in decreased numbers of T_{regs}, but not conventional CD4⁺ T cells (fig. S3), suggest-

ing that reduced IL-2R β signaling was selective for T_{regs}. With respect to other lymphoid cell populations, the proportions of CD19⁺ B cells, Ly6G⁺ granulocytes, CD11c⁺ dendritic cells, CD11b⁺ CD11c⁺ macrophages, and CD49b⁺ natural killer (NK) cells were similar between male NOD and NOD-Y3 mice (fig. S4), with the exception that the spleens of NOD-Y3 mice contained a reduced proportion of NK cells. These results, therefore, suggest that IL-2R β^{Y3} primarily affects T cells and that these changes are sufficient to accelerate the onset of diabetes.

IL-2R β^{Y3} increases the numbers of CD4⁺ Teff cells but decreases the numbers of CD8⁺ Teff cells in NOD-Y3 mice

To further evaluate the effects of IL-2R β^{Y3} on the conventional T cell compartment, we assessed cytokine production after stimulation of the T cells with phorbol 12-myristate 13-acetate (PMA) and ionomycin. Increased numbers of tumor necrosis factor- α (TNF- α)- and IL-2-producing CD4⁺ CD44^{hi} T cells were present in the PLNs of NOD-Y3 mice (Fig. 3D). These data associate the increased activity of CD4⁺ Teff cells with accelerated diabetes in NOD-Y3 mice. In contrast, a trend for reduced numbers of interferon- γ (IFN- γ), TNF- α , and IL-17⁺ CD8⁺ CD44^{hi} T cells was noted for NOD-Y3 mice, which was most obvious in the spleen (Fig. 3E). Thus, IL-2R β^{Y3} did not substantially alter the frequency of T helper 1 cells (T_H1) and T_H17 cells, which have been suggested to drive diabetes in NOD mice (35).

Reduced IL-2R signaling accounts for the altered numbers of CD4⁺ but not of CD8⁺ T cells in NOD-Y3 mice

The expression of IL-2R β^{Y3} resulted in a two- to threefold reduction in IL-15R signaling CD8⁺ T cells. IL-15 is important for the persistence of CD8⁺ T memory cells (36, 37). Thus, the lower proportion (Fig. 3C) and functional activity (Fig. 3E) of CD8⁺ T cells in NOD-Y3 mice might reflect impaired IL-15 signaling. Therefore, we also examined the effect of acute treatment with anti-IL-2 on WT NOD mice for

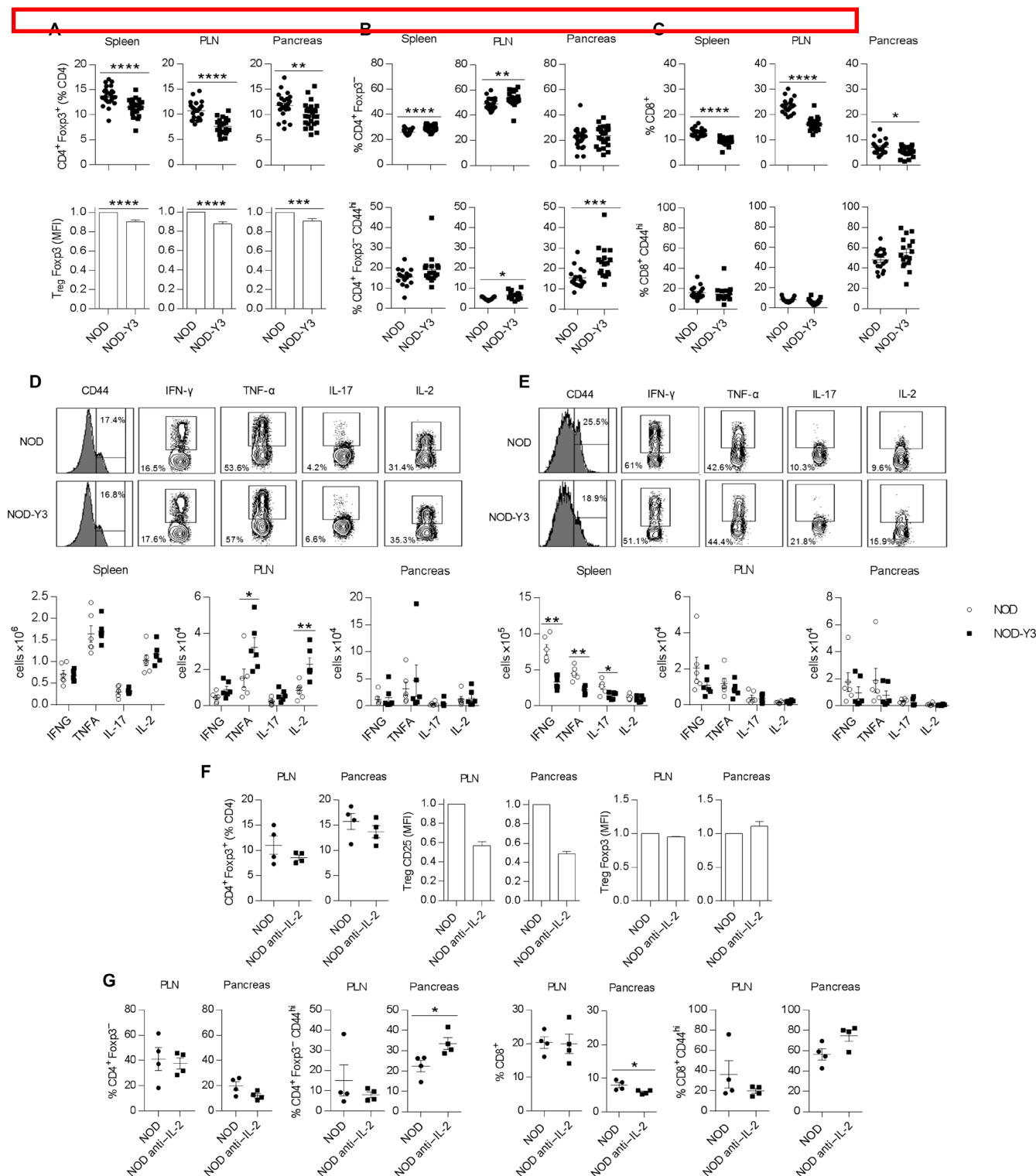


Fig. 3. The effect of IL-2R $\beta^{-/-}$ on T cells from male mice. (A to C) Flow cytometry analysis of the proportions of (A) Tregs and their amount of Foxp3, (B) total CD4⁺ Foxp3⁺ and CD4⁺ Foxp3⁺ CD44^{hi} antigen-experienced T cells, and (C) total CD8⁺ and CD8⁺ CD44^{hi} antigen-experienced T cells in the indicated tissues. Data are from 22 to 25 male mice per group and were analyzed by Mann-Whitney or Wilcoxon signed-rank tests for the MFIs. (D and E) Flow cytometry analysis of cytokine production by CD4⁺ and CD8⁺ effector T (Teff) cells. Representative flow cytometry profiles from the spleen (top) and cell numbers (bottom) of CD4⁺ CD44^{hi} cells (D) and CD8⁺ CD44^{hi} cells (E) from the indicated tissues that were positive for the indicated cytokines. Data were analyzed by Mann-Whitney test ($n = 6$). IFN- γ , interferon- γ ; TNF- α , tumor necrosis factor- α . (F and G) Analysis of the effect of anti-IL-2 on T cells in NOD mice. Mice received anti-IL-2 (S4B6) intraperitoneally, initially 1 mg on day 0 and 0.5 mg on days 2, 4, 7, and 9, and the indicated tissues were assessed 1 day after the last injection. (F) Effect on Tregs, including the amount of CD25 and Foxp3. (G) Effect on total CD4⁺ Foxp3⁺ and CD8⁺ T conventional cells and CD44^{hi} antigen-experienced CD4⁺ and CD8⁺ T cells. Data were analyzed by Mann-Whitney or Wilcoxon signed-rank tests for the MFIs ($n = 4$). * $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$, **** $P < 0.0001$.

CD4⁺T_{regs} and CD4⁺ and CD8⁺ T conventional cells. Similar to NOD-Y3 mice, anti-IL-2-treated NOD mice had reduced CD25 in T_{regs} and an increased proportion of antigen-experienced CD4⁺ Foxp3⁺ CD44^{hi} Foxp3⁺ T cells in the pancreas (Fig. 3, F and G). In contrast, anti-IL-2 had minimal effect on the overall proportion of CD8⁺ T cells and distribution of CD8⁺ antigen-experienced cells in NOD mice (Fig. 3G). Thus, these data are consistent with the notion that IL-2Rβ^{Y3} alters the CD4⁺ T_{reg} and CD4⁺ Foxp3⁺ antigen-experienced T cell compartments by reducing IL-2R signaling, whereas the reduction in the number of CD8⁺ T cells is likely a result of reduced IL-15R signaling.

T_{reg} subsets from IL-2Rβ^{Y3} mice have properties consistent with altered homeostasis

T_{regs} have been divided into subsets whereby increased cell surface expression of CD62L or CCR7 marks resting or central T_{regs} (cT_{regs}), whereas reduced expression of these molecules marks activated or effector T_{regs} (eT_{regs}) (11). Gene expression analysis indicates that eT_{regs} are heterogeneous; eT_{regs} that express CD69, CD103, or Klrp1 are more highly activated than their negative counterparts. In particular, these more activated eT_{regs} have increased amounts of T_{reg} functional molecules and chemokine receptors that promote cell trafficking into non-lymphoid tissues, among other changes (38, 39).

Because T_{regs} were consistently decreased in number in NOD-Y3 mice, including those that expressed the BDC2.5 TCR, we further focused on how IL-2Rβ^{Y3} might affect these cells. We assessed these T_{reg} subsets on the basis of their cell surface expression of CD62L and CD69 (Fig. 4A) and their expression of Ki67 (Fig. 4B) and Bcl-2 (Fig. 4C). This analysis revealed that NOD-Y3 mice had a reduced proportion of CD62L^{hi} CD69⁺ cT_{regs} compared to that of NOD mice (Fig. 4D). Accordingly, the spleen and pancreas contained an increased proportion of less activated CD69⁺ eT_{regs} (Fig. 4E), whereas the PLN contained more CD69⁺ eT_{regs} (Fig. 4F). This altered distribution of T_{reg} subsets in NOD-Y3 mice is accounted for in part by the altered homeostasis of these cells, as reflected by the increased abundance of Ki67 and the decreased abundance of Bcl-2. This phenotype was most apparent with cT_{regs} (Fig. 4D), which depend on IL-2 for their survival (11). However, these changes were also noted for CD69⁺ (Fig. 4E) and CD69⁺ (Fig. 4F) eT_{regs}. The increased abundance of Ki67 likely reflects the increased proliferation of T_{regs} to compensate for their impaired survival.

Transcriptional changes imposed by IL-2Rβ^{Y3} for T_{regs} from the pancreas were determined using the NanoString platform, which measures the expression of 547 immune-related genes. We found that 236 mRNAs were expressed by these T_{regs}, and 32 mRNAs were significantly differentially expressed ($P \leq 0.05$; table S1, $P < 0.05$). Consistent with our flow cytometry analysis, T_{regs} from NOD-Y3 mice expressed less *Il2ra* (CD25) and *Sell* (CD62L) mRNAs than did T_{regs} from NOD mice (Fig. 4G). A group of differentially expressed genes related to eT_{regs} exhibited decreased expression in IL-2Rβ^{Y3} T_{regs}, including *Irf4*, *Prdm1*, *Il10*, and *Cd247* (CD3ζ). Interferon regulatory factor 4 (IRF4) and TCR signaling are essential for eT_{reg} development (40, 41), whereas Blimp-1 promotes IL-10 production by eT_{regs} (40). Although *Il10* expression was decreased, gene expression of other T_{reg} functional molecules, such as *Ctla4*, *Ccr6*, *Entpd1*, *Nt5e*, and *Tgfb1*, were not differentially expressed between NOD and NOD-Y3 mice. Thus, these data suggest that, in addition to leading to altered T_{reg} homeostasis, IL-2Rβ^{Y3} also leads to a pancreatic T_{reg} signature of reduced functional activity due in part to the impaired development of the more highly activated, IL-10-producing eT_{regs}.

IL-2Rβ^{Y3} intrinsically alters the activity of CD4⁺ T_{regs}

The increase in the number of CD4⁺ Teff cells associated in NOD-Y3 mice might reflect a direct effect of impaired IL-2R signaling, or alternatively, it may be caused by an indirect effect due to an impaired T_{reg} compartment. To compare the diabetogenic capacity of autoantigen-specific CD4⁺ T cells, we transferred WT or IL-2Rβ^{Y3} CD4⁺ Foxp3⁺ BDC2.5 TCR transgenic T cells into NOD.SCID recipient mice to induce diabetes (Fig. 5A). We found that 50% of the recipient mice became diabetic between 13 and 16 days after the transfer, with a slight, but statistically nonsignificant delay, apparent in NOD.SCID recipient mice that received CD4⁺ T cells that expressed IL-2Rβ^{Y3}. These data suggest that IL-2Rβ^{Y3} does not accelerate diabetes by enhancing the function of autoreactive T cells.

To directly test the effect of IL-2Rβ^{Y3} on T_{reg} function, we cotransferred WT or IL-2Rβ^{Y3} BDC2.5 T_{regs} with WT BDC2.5 CD4⁺ Foxp3⁺ T cells into NOD.SCID recipients. In this case, only the recipients that received the WT BDC2.5 T_{regs} were protected from diabetes (Fig. 5B). For recipients that received BDC2.5 IL-2Rβ^{Y3} T_{regs}, donor WT BDC2.5 CD4⁺ Foxp3⁺ T cells minimally increased in number in the spleen but substantially increased in the pancreas because of increased proliferation as determined by measuring Ki67 abundance (Fig. 5C). Furthermore, NOD-Y3 BDC2.5 T_{regs} were found at a much lower proportion than NOD BDC2.5 T_{regs}, consistent with an inability to suppress the autoreactive BDC2.5 T cells (Fig. 5D). These data suggest that IL-2Rβ^{Y3} directly dysregulates T_{reg} function.

IL-2Rβ^{Y3} impairs the homeostasis of T_{reg} subsets

To assess the possibility that altered T_{reg} homeostasis also contributed to the acceleration of diabetes in NOD-Y3 mice, mixed bone marrow chimeric mice were prepared such that irradiated NOD.SCID recipients received a 1:1 mixture of T cell-depleted bone marrow from NOD and NOD-Y3 mice (Fig. 6A). Eight weeks after transplantation, the distribution of donor T cells was assessed for the spleen, PLN, and pancreas (Fig. 6B). T_{regs} of IL-2Rβ^{Y3} origin were substantially underrepresented in these recipients, whereas equivalent proportions of CD4⁺ Foxp3⁺ and CD8⁺ T cells of WT and IL-2Rβ^{Y3} origin were found in these tissues (Fig. 6C). In addition, IL-2Rβ^{Y3} T_{regs} had decreased amounts of CD25 and Foxp3, consistent with reduced IL-2R signaling, and had decreased Ki67 abundance, consistent with impaired proliferation (Fig. 6D).

Although cT_{regs} were reduced in number in NOD-Y3 mice compared to those in NOD mice (Fig. 4D), the proportions of cT_{regs} and eT_{regs} did not substantially vary between T_{regs} of NOD and NOD-Y3 origin in the bone marrow chimeras (Fig. 6E). However, the number of NOD-Y3 eT_{regs} that expressed CD103, inducible T cell costimulator (ICOS), and Klrp1 was statistically significantly reduced, consistent with the impaired development of highly activated and terminally differentiated T_{regs}. If cell numbers rather than percentages are considered, then the numbers of IL-2Rβ^{Y3} cT_{regs} and most eT_{reg} subsets were markedly reduced (Fig. 6F). These data indicate that IL-2Rβ^{Y3} cT_{regs} did not effectively compete with WT NOD cT_{regs}, consistent with the impaired homeostasis of this T_{reg} subset, and suggest that IL-2Rβ^{Y3} T_{regs} cannot properly develop into highly activated subsets.

IL-2Rβ^{Y3} impairs the development of eT_{reg} subsets

To directly assess the development of eT_{regs} from cT_{regs}, CD62L^{hi} cT_{regs} were isolated from NOD and NOD-Y3 mice and transferred at a 1:1 ratio into NOD-Y3 IL-2Rβ^{Y3} recipient mice (Fig. 7A). We have previously used this type of recipients, but on the C57BL/6 genetic background, to

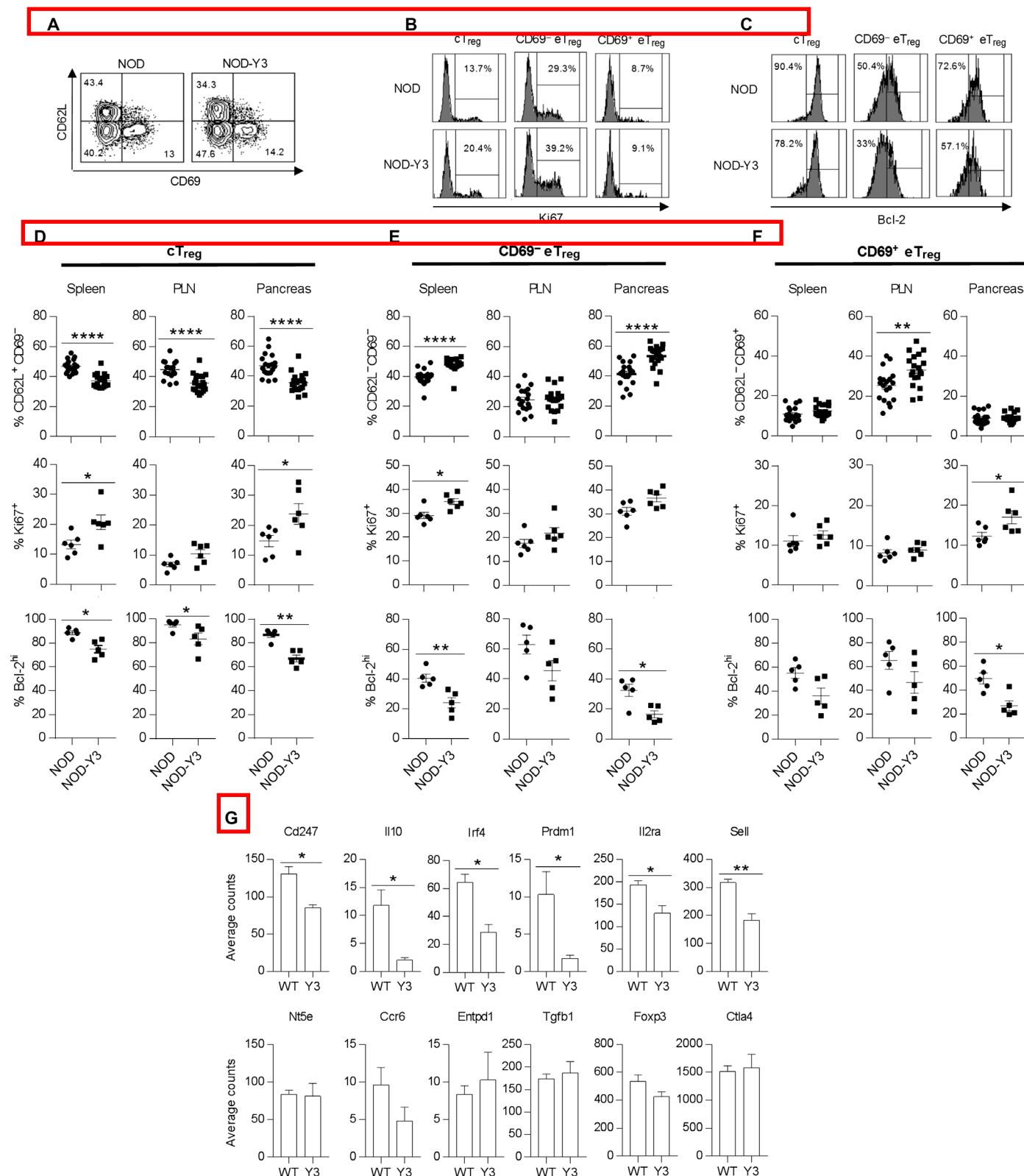


Fig. 4. Altered distribution of T_{reg} subsets in male NOD-Y3 mice. (A to C) Representative flow cytometry plots of T_{reg} subsets from the spleens of male NOD and NOD-Y3 mice based on the expression of CD62L and CD69 (A), T_{reg} subset Ki67 (B), or T_{reg} subset Bcl-2 (C). (D to F) Analysis of the percentages of the designated subsets ($n = 19$) and their expression of Ki67 ($n = 6$) and Bcl-2 ($n = 5$) for central T_{reg}s (cT_{reg}s) (D), CD69⁻ effector T_{reg}s (eT_{reg}s) (E), and CD69⁺ eT_{reg}s (F). Data were analyzed by Mann-Whitney test. (G) The abundances of the indicated mRNAs in pancreatic T_{reg}s from male NOD and NOD-Y3 mice were assessed with the NanoString platform. Data were analyzed by two-tailed unpaired t test with the nSolver analysis software package from NanoString ($n = 3$). WT, wild-type. * $P < 0.05$, ** $P < 0.01$, **** $P < 0.0001$.

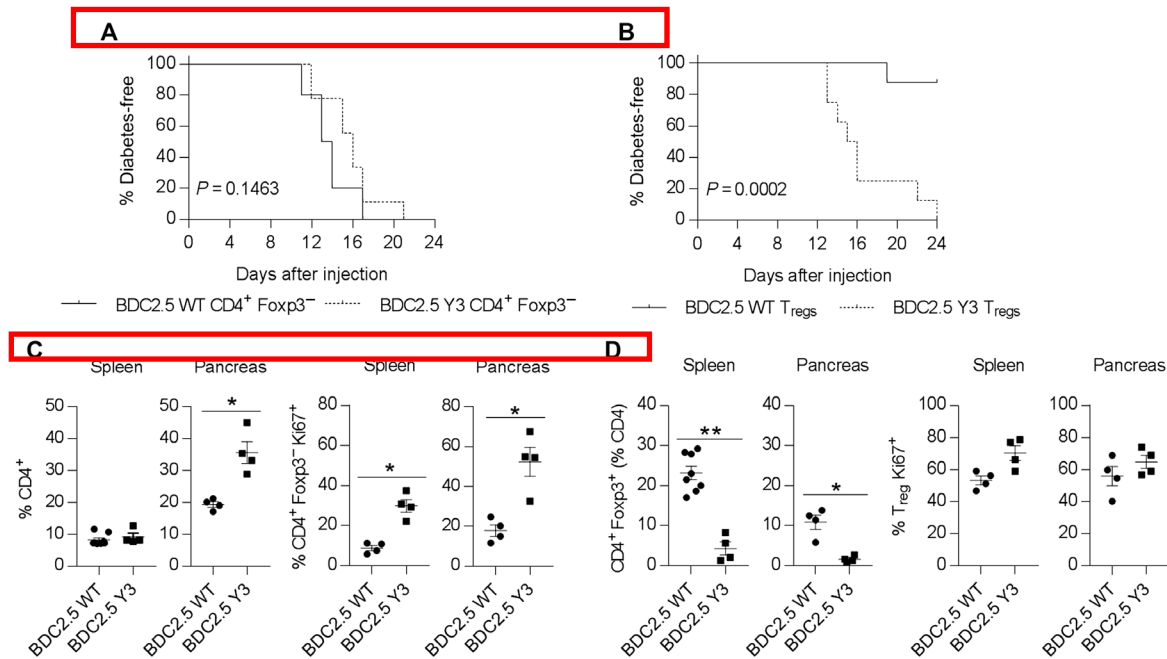


Fig. 5. Functional activity of Teff and T_{reg} cells that express IL-2R β ^{Y3} on the NOD BDC2.5 background. **(A)** Analysis of the onset of diabetes induced by CD4⁺ Teff cells. CD4⁺ Foxp3⁻ T cells (5×10^4) from NOD BDC2.5 or NOD-Y3 BDC2.5 mice were transferred into NOD.SCID recipients, which were monitored for 24 days. **(B)** Analysis of the suppression of diabetes by T_{regs}. Purified WT or IL-2R β ^{Y3} BDC2.5 T_{regs} (2.5×10^4) were cotransferred with WT CD4⁺ Foxp3⁻ BDC2.5 T cells (5×10^4) into NOD.SCID mice, which were monitored for 24 days. Data in (A) and (B) were analyzed by log-rank test ($n = 9$ to 10). **(C and D)** Analysis of the properties of Teff cells (C) and T_{regs} (D) from the mice in (B), where BDC2.5-WT or BDC2.5-Y3 on the x axis refer to the type of T_{regs} transferred into the NOD.SCID recipients. Mice were analyzed when the mice were determined to be diabetic or 24 days after transfer. Data were analyzed by Mann-Whitney test ($n = 4$ to 8). * $P < 0.05$, ** $P < 0.01$.

study T_{reg} subset development (38). T_{regs} from C57BL/6-Y3 IL-2R β ^{-/-} mice have a more substantial block in IL-2R β signaling than T_{regs} from C57BL/6-Y3 IL-2R β ^{+/+} mice (32), but they are not lymphopenic. These properties were also found for NOD-Y3 IL-2R β ^{+/+} and NOD-Y3 IL-2R β ^{-/-} mice (Fig. 1C). When C57BL/6-Y3 IL-2R β ^{-/-} mice were used as recipients, reduced IL-2R β signaling associated with recipient-derived T_{regs} was favorable for engraftment of relatively small numbers of donor cT_{regs}. This environment makes it feasible to study the development of T_{reg} subpopulations in vivo without any conditioning regimen to make space for donor T_{regs}.

We purified cT_{regs} based on CD4⁺ T cells that coexpressed CD25 and CD62L (Fig. 7B). Donor WT cells were >93% Foxp3⁺. Engrafted WT donor T_{regs} were identified solely on the basis of their expression of CD45.2, whereas donor IL-2R β ^{Y3} T_{regs} were identified on the basis of their expression of CD45.1 and red fluorescent protein (RFP; Fig. 7C). These cells were distinguished from recipient T_{regs}, which were CD45.1⁺ but RFP⁻. This analysis revealed that the persistent engrafted WT and IL-2R β ^{Y3} donor cells were 95% Foxp3⁺. However, we cannot assess the fraction, if any, of donor IL-2R β ^{Y3} T_{regs} that may have converted to Foxp3⁻ Teff cells.

Two weeks after transfer, the donor T_{regs} were about 80% of WT (CD45.2⁺) origin in the spleen, PLN, and pancreas (Fig. 7D). T_{regs} of NOD-Y3 origin expressed less CD25 and CD62L, the latter indicative of a reduced proportion of cT_{regs} (Fig. 7D). These data further demonstrate the impaired homeostasis of cT_{regs} with reduced IL-2R β signaling associated with IL-2R β ^{Y3}. Furthermore, the development of eT_{regs} was also abnormal. Compared to WT T_{regs}, IL-2R β ^{Y3} T_{regs} had reduced amounts of ICOS, CD103, and Klrp1 (Fig. 7D). These data indicate that IL-2R β ^{Y3} T_{regs} are impaired in their ability to generate highly activated eT_{reg} subsets. Together, these data suggest that reducing IL-2R signal-

ing in T_{regs} leads to compromised homeostasis of cT_{regs} and the altered development of eT_{regs}.

DISCUSSION

Past studies provided some initial information consistent with the notion that variation in the genes encoding IL-2 or the IL-2R in mice and humans reduces the activity of T_{regs} (21, 24, 26), but little is understood regarding how this genetic variation represents a risk for autoimmunity. Here, we directly reduced IL-2R β signaling in T cells and showed that modestly, but chronically, reducing IL-2 responsiveness by about 2.6-fold accelerated diabetes in male and female NOD mice. Our findings demonstrate that accelerated diabetes occurs because of a multiplicity of effects on T_{regs} that affect their homeostasis and function. Altered T_{reg} homeostasis by IL-2R β ^{Y3} was evident because T_{regs} from NOD-Y3 mice were broadly characterized by having increased amounts of Ki67 but decreased Bcl-2 abundance. This phenotype is consistent with T_{regs} that undergo increased proliferation to counterbalance enhanced cell death. Reduced IL-2R signaling affected T_{reg} homeostasis in two distinct ways. First, cT_{regs} (as defined by high expression of CD62L) were found at a reduced proportion under several different experimental conditions when they expressed IL-2R β ^{Y3}, and this effect may reflect impaired survival because IL-2 promotes the survival of this T_{reg} subset (11). Second, the development of more highly activated eT_{regs} [identified on the basis of the expression of CD103 and Klrp1 by CD62L^{lo} T_{regs} (38)] was less efficient for T_{regs} with reduced IL-2R signaling. This was most obviously seen in experiments with mixed bone marrow chimeras and the direct development of eT_{regs} from purified adoptively transferred cT_{regs}. The expression of Klrp1 by eT_{regs} represents a terminally differentiated state of activation and occurs only after

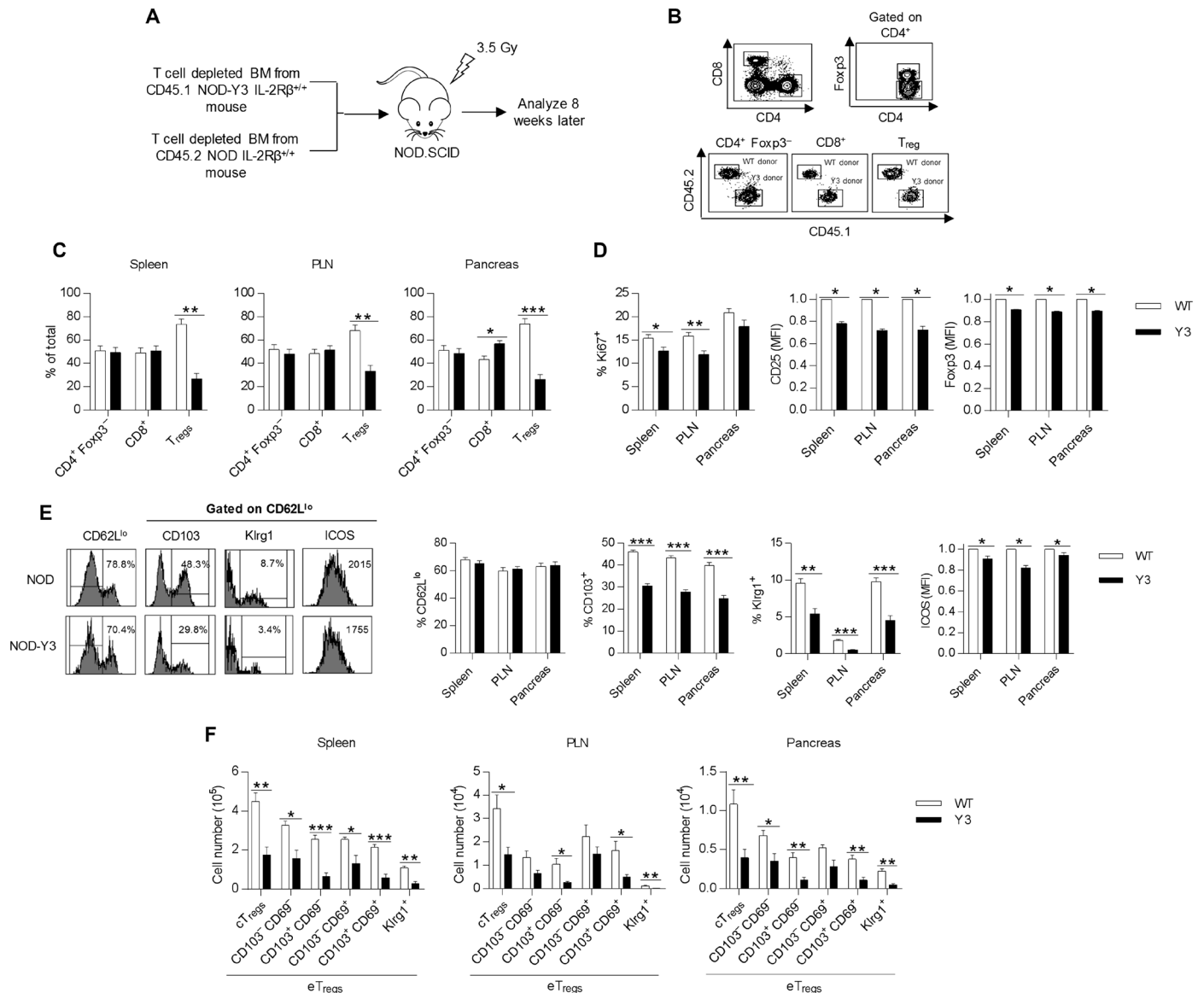


Fig. 6. Selective inability of IL-2R β^{Y3} Tregs to compete with WT Tregs in vivo. (A) Experimental scheme for establishing the mixed bone marrow (BM) chimeras. (B) Representative flow cytometry plots of the donor cell populations that were isolated from NOD.SCID recipient mice 8 weeks after transfer. (C) Percentages of the donor cells in the indicated tissues. Data were analyzed by Mann-Whitney test ($n = 7$). (D) Flow cytometry analysis of the expression of Ki67, CD25, and Foxp3 in Tregs from the indicated tissues. Data analyzed by Mann-Whitney test and Wilcoxon signed-rank test for MFI ($n = 7$). (E) Representative flow plots from the spleen (left) and analysis of the percentage of CD62L lo Tregs and their expression of CD103, Klr1, and ICOS (right) from the indicated tissues. Data for ICOS (MFI) were analyzed by Wilcoxon signed-rank test; the remaining data were analyzed by Mann-Whitney test ($n = 7$). (F) Analysis of the numbers of the indicated Treg subsets from the indicated tissues based on the expression of CD62L, CD69, CD103, and Klr1. Data were analyzed by Mann-Whitney test ($n = 7$). * $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$.

more than eight or nine cell divisions as cTregs develop into eTregs (12). The observation that reduced IL-2R signaling led to fewer Klr1 $^{+}$ eTregs is likely explained by eTregs that die before reaching this state of activation.

In other systems, blockade of IL-2R signaling destabilizes Tregs so that they become IFN- γ^{+} and IL-17 $^{+}$ Teff cells; this process is most evident for peripherally derived Tregs (42, 43). Our experimental design did not enable us to directly assess this process, and some of the decrease in Treg number that we noted might be due to their development into Teff cells. However, several observations suggest that this mechanism does not fully account for the accelerated onset of diabetes in

NOD-Y3 mice. First, we did not detect increases in the numbers of IFN- γ - and IL-17-producing CD4 $^{+}$ T cells in NOD-Y3 mice. If Treg conversion to T H 1 and T H 17 Teff cells was a main mechanism, these cells should have increased in number, particularly in the pancreas. Second, TCR repertoire studies indicate that most Tregs in the peripheral immune tissues are thymus-derived cells (44) and have a stable phenotype (45). Thus, the poor persistence of adoptively transferred IL-2R β^{Y3} Tregs is unlikely to be due to the few peripherally derived Tregs that may be part of the donor inoculum.

This study also begins to provide a molecular underpinning for the dysregulation associated with IL-2R β^{Y3} Tregs. IRF4 is required for the

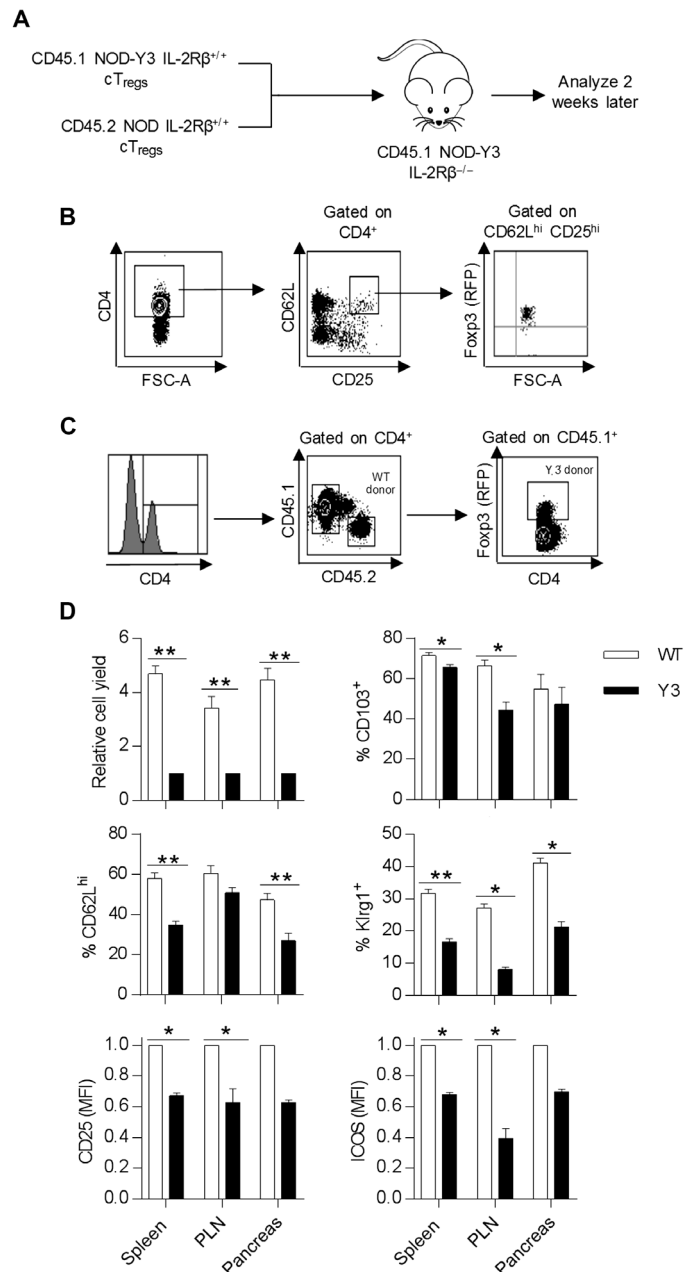


Fig. 7. IL-2Rβ^{Y3} cTregs show impaired development into activated eTreg subsets.

(A) Experimental scheme showing how purified T_{regs} (8×10^4 to 15×10^4) from NOD and NOD-Y3 mice were premixed at a 1:1 ratio and verified by flow cytometry analysis before being transferred to NOD-Y3 IL-2Rβ^{-/-} recipients. FCS-A, forward light scatter-parameter A. **(B)** Representative sorting strategy to purify donor-derived cT_{regs}. RFP, red fluorescent protein. **(C)** Representative flow plots to identify donor populations in male NOD-Y3 IL-2Rβ^{-/-} hosts 2 weeks after transfer. **(D)** Relative donor T_{reg} yields from the indicated tissues ($n = 5$ to 6) and their expression of CD62L ($n = 5$ to 6), CD25 ($n = 5$ to 6), CD103 ($n = 3$ to 4), Klr1g1 ($n = 4$ to 6), and ICOS ($n = 5$ to 6) from the indicated tissues. Data were analyzed by Wilcoxon signed-rank test for MFIs, whereas the Mann-Whitney test was used for the remaining samples. * $P < 0.05$, ** $P < 0.01$.

cT_{reg}-to-eT_{reg} developmental step (40). Blimp-1 is selectively expressed by eT_{regs} and contributes to the production of IL-10 (40). Broad immune mRNA profiling by NanoString analysis showed that all of these mRNAs were less abundant in pancreatic T_{regs} of NOD-Y3 mice than

in those of WT mice. These data are consistent with reduced IL-2R signaling interfering with the development of eT_{regs} through an IRF4–Blimp-1–IL-10 axis. Activated STAT5 directly regulates the expression of *Il2ra*, *Foxp3*, *Irf4*, and *Prdm1* (46–48). In addition, *Il2ra*, *Irf4*, and *Prdm1* are direct targets of Foxp3 (49). Thus, subtly and chronically reducing both IL-2R signaling and the abundance of Foxp3 by IL-2Rβ^{Y3} might cooperatively act to impair eT_{reg} development. TCR signaling also importantly contributes to the development of eT_{regs} (41). Thus, the reduced abundance of CD3ζ in pancreatic IL-2Rβ^{Y3} T_{regs} raises the possibility that IL-2Rβ^{Y3} might also influence eT_{reg} development by reducing TCR signaling, but this requires further investigation.

We showed that IL-2Rβ^{Y3} directly reduced functional activity by demonstrating the inability of T_{regs} from NOD-Y3 BDC2.5 mice to suppress autoreactive BDC2.5 T cells and diabetes development in a NOD.SCID transfer model. The effect of IL-2Rβ^{Y3} on T_{regs} is predicted to reduce T_{reg} function in several important ways. The impaired homeostasis of cT_{regs} and eT_{regs} likely diminishes the capacity of T_{regs} that reside in lymph nodes and tissue sites, respectively, to properly balance their numbers with autoreactive T cells. Furthermore, Klr1g1⁺ and CD103⁺ eT_{regs} are characterized by having increased amounts of several T_{reg} functional molecules and tissue-targeting chemokine receptors (38, 39). Thus, IL-2-dependent alterations in eT_{reg} development affect T_{reg} function indirectly by reducing the proportions of T_{regs} with enhanced suppressive function, as well as directly by decreasing the amount of IL-10.

Previous studies designed to ascertain the contribution of IL-2R signaling to diabetes in NOD mice have been indirect in that they assessed the potential effects of reduced IL-2 production, especially by pancreatic T cells, on T_{regs} due to the *Idd3* polymorphism (26). These studies implied that NOD T_{reg} homeostasis was impaired as reflected by their altered amounts of Ki67 and Bcl-2, which we confirmed. A complication of the past results is that *Idd3* not only reduces IL-2 but also increases the production of the proinflammatory cytokine IL-21 (28), which might also contribute to the development of diabetes. Thus, an advantage of IL-2Rβ^{Y3} is that IL-2R signaling is directly targeted. The caveat with IL-2Rβ^{Y3} is that IL-15R signaling is also reduced. This was detected in T_{regs} and CD8⁺ T cells. However, the T_{reg} compartment is largely normal in IL-15-deficient mice (6), and we showed that in situ pSTAT5 activation by T_{regs} was inhibited by anti-IL-2, indicating that impaired IL-2R signaling largely accounted for the intrinsic effects of IL-2Rβ^{Y3} on T_{regs}. In contrast, the lower proportion of CD8⁺ T cells associated with NOD-Y3 mice was unaffected by anti-IL-2. These results, in conjunction with the known role of IL-15 in CD8⁺ T cell homeostasis (50, 51), suggest that the decrease in the number of CD8⁺ T cells is accounted for by reduced IL-15R signaling. CD8⁺ T cells and increased IL-15 signaling have been implicated in promoting diabetes in NOD mice (52). Thus, a reduction in these two parameters by IL-2Rβ^{Y3} is not expected to be a contributing factor to the accelerated disease in NOD-Y3 mice.

Upon reducing IL-2R signaling in T cells, T_{regs} were intrinsically affected in a manner that would reduce peripheral tolerance, as discussed earlier. Reduced IL-2R signaling is also expected to reduce T_H1 and T_H2 responses (53–56), which should not accelerate diabetes. However, decreased IL-2R signaling could intrinsically increase the concentration of IL-2, T_H17s, and T follicular helper (T_{fh}) cells (57, 58), whereas increased IL-17 has been linked to diabetes in NOD mice (59). We observed an increase in antigen-experienced CD4⁺ T cells in NOD-Y3 mice, which included increased numbers of IL-2-producing, but not IL-17-producing, T cells in the PLN. Increased anti-insulin

autoantibodies were also observed, which may reflect an increase in the number of T_{fh} cells due to reduced STAT5 and Blimp-1-mediated repression (57). However, transfer studies of CD4⁺ Foxp3⁺ T cells into NOD.SCID mice did not provide clear evidence that IL-2Rβ^{Y3} enhanced the activity of autoreactive T cells. A more difficult question to resolve is whether accelerated diabetes in NOD-Y3 mice is due to the reduced intrinsic IL-2Rβ signaling by autoreactive T cells that enhances their activity or is simply the result of reduced T_{reg} activity.

Although male NOD mice are relatively resistant to diabetes, a particularly marked result was that NOD-Y3 males developed diabetes at a similar rate to that of NOD females. Sex-biased development of T1D in NOD mice has been attributed to differences in microbiota and sex hormones (60–62). Differences in sex hormones have been suggested to influence the type of microbiota in male and female NOD mice (62), raising the possibility that these two factors are interrelated. Although our study did not address the factors responsible for the lack of sex bias in NOD-Y3 mice, we speculate that altering the number of eT_{regs} (which dominate the gut mucosa) through reduced IL-2R signaling distinctively shapes immune homeostasis in the gut mucosa to influence the microbiota in a way that promotes diabetes in both male and female mice. Alternatively, decreased T_{reg} function as a result of reduced IL-2R signaling in NOD-Y3 mice may be sufficient to counteract the protective effects in male NOD mice that are attributed to microbiota and sex hormones. Experimentation is required to assess these points.

The IL-2–IL-2R axis has been implicated as a risk factor not only for T1D but also for other autoimmune diseases, including multiple sclerosis, celiac disease, and rheumatoid arthritis. IL-2Rβ^{Y3} reduced IL-2R signaling in all T cells, and this situation is predicted to mimic genetic variation due to SNPs within the noncoding region of *IL2RA* that may alter gene expression. Thus, the effects that we noted in NOD-Y3 mice might also be detected in humans that carry some *IL2RA* risk alleles. In addition, low-dose IL-2 represents a promising strategy to boost T_{regs} in patients with autoimmunity (63). Our study raises the possibility that along with evaluating the capacity of low-dose IL-2 to increase the total number of T_{regs}, one may also wish to assess how this treatment affects particular subsets of T_{regs} to maximize therapeutic benefits.

MATERIALS AND METHODS

Mice

NOD CD45.2 congenic (*NOD.B6-Ptprc^b/6908MrkJacJ*) and NOD.SCID (*NOD.CB17-Prkdc^{SCID}/J*) mice were purchased from Jackson Laboratories. NOD-Y3 IL-2Rβ^{+/+}, NOD-Y3 IL-2Rβ^{+/-}, and NOD-Y3 IL-2Rβ^{-/-} mice, where Y3 refers to the mutant IL-2Rβ transgene, were developed by backcrossing C57BL/6-Y3 IL-2Rβ^{-/-} mice (32) to *NOD/ShiLtJ* mice for 12 generations. All *Idd* loci were confirmed to be of NOD origin. NOD/Foxp3-RFP reporter (FIR) mice were previously described (64) and crossed to NOD BDC2.5 and NOD-Y3 IL-2Rβ^{+/+} mice to yield NOD-Y3-FIR, NOD BDC2.5-FIR, and NOD-Y3 BDC2.5-FIR mice. All mice were housed in a specific pathogen-free animal facility at the University of Miami. The Institutional Animal Care and Use Committee at the University of Miami reviewed and approved all animal studies conducted in this study.

Assessment of diabetes, insulin autoantibodies, and islet infiltration scoring

Diabetes was determined by following urine glucose two to three times per week using Diastix strips (Bayer). After two positive urine tests

(>250 mg/dl), blood glucose was assessed using OneTouch Ultra strips. Mice were considered diabetic when blood glucose readings were >250 mg/dl. Serum was analyzed for insulin autoantibodies by radioimmunoassay as previously described (65). To assess islet inflammation, the pancreas was fixed in 10% neutral buffered formalin, embedded in paraffin, cut into 5-μm sections, and stained with hematoxylin and eosin. Lymphocytic infiltration was scored for each islet and assessed blindly by light microscopy. Insulitis scoring was conducted per the following criteria: 0, no insulitis (free of infiltration); 1, peri/polar insulitis (infiltration confined to the periphery of islets); 2, mild insulitis (<50% of the islet area infiltrated); 3, severe insulitis (≥50% of the islet area infiltrated); and 4, massive insulitis (≥90% of the islet area infiltrated) (66).

Fluorescence-activated cell sorting analysis

Cell staining for fluorescence-activated cell sorter (FACS) analysis was performed as previously described (67). Monoclonal antibodies (mAbs) specific for the following targets were used for flow cytometry: CD25 (PC61), pSTAT5 (pY641), Ki67 (B56), and CD11c (HL3) were purchased from BD Biosciences; and Foxp3 (FJK-16s), CD62L (MEL-14), IL-2 (JES6-5H4), CD103 (2E7), Klrq1 (2F1), ICOS (15F9), and streptavidin were purchased from eBioscience. Antibodies against CD122 (Tm-b1), CD4 (RM4-5), CD8 (53-6.7), IFN-γ (XMG1.2), TNF-α (MPG-XT22), IL-17 (TC11-18H10.1), CD69 (H1.2F3), Bcl-2 (BCL10C4), CD45.1 (A20), CD45.2 (104), CD19 (GD5), CD49b (DX5), Ly6G (1A8), CD11b (M1/70), and streptavidin were purchased from BioLegend. Antibodies against CD4 (GK1.5) and CD44 (PGP-1) were purified and conjugated in-house. Intracellular staining for Foxp3 (eBioscience) was performed as recommended by the manufacturer. For intracellular staining of cytokines, the BD Cytofix/Cytoperm kit was used according to the manufacturer's protocol. Samples were analyzed on a BD LSRFortessa-HTS or a BD LSR II flow cytometer. Typically, 200,000 total events were collected per sample. Flow cytometry data were analyzed with BD FACSDiva software (version 8.0.1).

Cell isolation and sorting

To isolate lymphoid cells from the mouse pancreas, pancreases were minced into about 2-mm² pieces in Hanks' balanced salt solution (HBSS) containing 50% fetal calf serum (FCS) and incubated at 37°C for 3 hours in a 7% CO₂ incubator. The total pancreas pieces were then passed through a 70-μm nylon strainer (Falcon). The resulting single-cell suspension was centrifuged at 480g for 5 min, pelleted, and then washed once with HBSS. To prepare T cell-depleted bone marrow cells, mAbs against CD4 (0.2 ml of culture supernatant; RL1724), CD8 (0.2 ml of culture supernatant; HO22), Thy-1.2 (10 μg of purified antibody; 30-H12, Sigma-Aldrich), deoxyribonuclease (10 μg; Sigma-Aldrich), and rabbit complement (0.12 ml; Cedarlane Laboratories) were added to 1 ml of bone marrow cells (25 × 10⁶/ml in RPMI 1640 containing 5% FCS). This mixture was incubated on ice for 15 min and then incubated at 37°C for 30 min. After this time, the cells were washed once in RPMI 1640. In most experiments, CD4⁺ T_{regs} and T conventional cells from the spleen or pancreas were surface-stained and sorted on the basis of the presence or absence, respectively, of the Foxp3-RFP reporter with FACSaria II or Beckman Coulter MoFlo Astrios EQ cell sorters. Cell purity was usually greater than 95%. For the adoptive transfer of T_{regs} into NOD-Y3 IL-2Rβ^{-/-} mice, where some donor mice were not on the Foxp3-RFP reporter background, T_{regs} were sorted on the basis of CD4⁺ CD62L^{hi} CD25^{hi} expression. This population was greater than 93% Foxp3⁺ cells.

Functional and molecular analyses

To measure intracellular cytokines by flow cytometry, cells (1×10^6 to 2×10^6) were cultured in 24-well flat-bottom plates with RPMI 1640 complete medium (68) containing PMA (50 ng/ml), ionomycin (1 μ M), and brefeldin A (5 μ g; BioLegend) for 4 hours. To assess mRNA abundance, cell lysates from FACS-purified cells were prepared with RLT lysis buffer (Qiagen). These lysates (10,000 cell equivalents) were subjected to NanoString analysis using the nCounter Mouse Immunology Gene Expression Code set (NanoString). Expression data were analyzed by normalizing counts to housekeeping genes and positive and negative controls using nSolver version 2.5 (38). Normalized counts were $\log_2$ -transformed, and the data were analyzed by Student's two-tailed *t* test. IL-2-dependent (eBioscience) and IL-15-dependent (PeproTech) pSTAT5 activation was assessed by flow cytometry as previously described (7), either directly ex vivo or after stimulation in vitro with the appropriate cytokine for 15 min. The resulting dose-response curve data were subjected to nonlinear regression analysis using GraphPad Prism software to determine the EC_{50} .

In vivo studies

For adoptive transfer studies, the appropriate cell populations were transferred intravenously through the tail vein into NOD.SCID or NOD-Y3 IL-2R $\beta^{-/-}$ recipient mice. Glucose concentrations were monitored in the NOD.SCID recipients for up to 24 days after transfer, as required. For bone marrow chimeras, 1 day before cell transfer, NOD.SCID recipients were subjected to 3.5 Gy of γ irradiation. Donor T cell-depleted bone marrow was mixed, as indicated in Results, and injected intravenously into the tail vein. Blood was collected to monitor donor cell reconstitution before the recipient mice were sacrificed 8 weeks after transfer.

Statistical analysis

All data are represented as means $\pm$ SEM. Statistical tests used are listed in the figure legends. Data were analyzed using Wilcoxon signed-rank tests for FACS expression based on the MFI, where the MFI of cells from WT NOD mice were normalized to 1. The log-rank test was used to assess diabetes instances, the Kruskal-Wallis test was used to assess multigroup comparisons, and the Mann-Whitney test was used to assess autoantibody titers and two group comparisons. Two-tailed unpaired *t* tests were used to analyze pancreatic T_{reg} NanoString data. Statistical comparisons were performed using GraphPad Prism (version 7.01). Statistical significance is indicated in the figure legends as follows: **P* < 0.05, ***P* < 0.01, ****P* < 0.001, *****P* < 0.0001.

SUPPLEMENTARY MATERIALS

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Fig. S1. Analysis of STAT5 activation in conventional T cells from male NOD-Y3 mice.

Fig. S2. Flow cytometry analysis of T lymphocytes from female NOD-Y3 mice.

Fig. S3. Analysis of IL-2R β^{Y3} expression on T lymphocytes from mice on the NOD BDC2.5 background.

Fig. S4. Analysis of cell distribution in the non-T cell compartment in male NOD-Y3 mice.

Table S1. NanoString analysis of mRNA expression in pancreatic T_{regs} from male NOD and NOD-Y3 mice.

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Acknowledgments: We thank the flow cytometry cores of the Diabetes Research Institute and the Sylvester Comprehensive Cancer Center at the University of Miami for help with cell sorting and flow cytometry analysis, K. Johnson for preparation of pancreatic sections, and the Onco-Genomics Core Facility at the University of Miami for help with NanoString sample preparation. **Funding:** This research was supported by the NIH grant R01DK093866 and the Diabetes Research Foundation, Hollywood, Florida. **Author contributions:** T.R.M. conceived and supervised the study. T.R.M., Z.C., and C.J.D. designed experiments. A.L.B. and C.C.-K. developed the NOD-Y3 mice. C.J.D. performed most of the experiments with help from C.F. in assessing islet inflammation, L.Y. in determining the insulin autoantibody titers, N.C.W. with the anti-IL-2 in vivo experiments, and K.H.T. in assessing the development of eT_{regs} from cT_{regs}. C.J.D. and T.R.M. analyzed the data. C.J.D. and T.R.M. wrote the paper with edits from A.L.B., C.F., K.H.T., and Z.C. **Competing interests:** The authors declare that they have no competing interests.

Submitted 10 February 2017
Accepted 1 December 2017
Published 19 December 2017
10.1126/scisignal.aam9563

Citation: C. J. Dwyer, A. L. Bayer, C. Fotino, L. Yu, C. Cabello-Kindelan, N. C. Ward, K. H. Toomer, Z. Chen, T. R. Malek, Altered homeostasis and development of regulatory T cell subsets represent an IL-2R-dependent risk for diabetes in NOD mice. *Sci. Signal.* **10**, eaam9563 (2017).

Altered homeostasis and development of regulatory T cell subsets represent an IL-2R–dependent risk for diabetes in NOD mice

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Sci. Signal. **10** (510), eaam9563.
DOI: 10.1126/scisignal.aam9563

IL-2 signaling and autoimmunity

Regulatory T cells (T_{regs}) suppress autoreactive effector T cells to prevent the occurrence of autoimmune diseases, such as type 1 diabetes (T1D). The cytokine interleukin-2 (IL-2) is critical for the development and homeostasis of T_{reg} subsets. Polymorphisms in the genes encoding IL-2 and its receptor subunits are associated with an increased risk of developing autoimmunity. To examine the effect of decreased IL-2 signaling, Dwyer *et al.* expressed a signaling-defective mutant IL-2 receptor (IL-2R β^{Y3}) in T cells in NOD mice, a model of T1D. Compared to NOD mice expressing wild-type IL-2R β , those expressing IL-2R β^{Y3} in their T cells had accelerated onset of T1D. This was associated with a decrease in the numbers and suppressive activity of different T_{reg} subsets and in the infiltration of autoreactive effector T cells into the pancreas. Together, these data suggest that the use of low-dose IL-2 to therapeutically modulate different T_{reg} subsets in the context of autoimmune disease should be evaluated.

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